

David Abella Bujalance

Physicist. Researcher in Computational Social Science.

Barcelona, Spain

davidabbu.github.io

davidabbu@gmail.com

Skills

Latex

Office

Fortran

Julia

Python

(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Languages

- Catalan - Native language
- Spanish - Native language
- English - Advanced level

Interests

As a physicist, my expertise lies in statistical mechanics, network science and data analysis, a powerful toolkit to decipher complex problems. I apply these skills to social and urban challenges, such as housing markets, urban segregation, collective behavior and the governance of online platforms like Wikipedia.

Work Experience

2025-Now — Researcher in Computational Social Science at Eurecat

Eurecat, Centre Tecnològic de Catalunya. Projects: DEM-Debate (Wikipedia editing and content moderation during the 2024 European Parliament election, with Wikimedia Europe and IViR, University of Amsterdam); AI4Bus (AI forecasting of passenger flows for Barcelona's metropolitan bus network, with AMB); deployEMDS (deployment of the common European Mobility Data Space).

2023-2024 — Research Technician at IFISC (CSIC)

Research associated with the projects Nou Lloguer and Next4Mob to continue renting market data analysis and modelling.

2022-2023 — Young researchers scholarship by SOIB (UIB)

Research focused on different agent-based models of human dynamics.

2022 — Research Technician at IFISC (CSIC)

Research associated with the projects APASOS-CSIC and Nou Lloguer for the renting market data analysis.

2021 — Predoctoral contract at IFISC

8-month contract to continue the study of aging effects in threshold models with application in sociotechnical systems. Focused on the study of aging in the Noisy Schelling segregation model.

Education

2021-2024 — PhD Physics of Complex Systems at the Institute for Cross-Disciplinary Physics and Complex Systems (IFISC, UIB-CSIC)

Thesis: *Aging and memory effects in social and economic dynamics*. Supervisors: Maxi San Miguel and José J. Ramasco. Awarded the doctoral thesis prize of the UIB Chair of Urban Studies (2024).

2019-2020 — M.Sc Physics of Complex Systems at Universitat de les Illes Balears

Complex systems and computational social science, applying physics to interdisciplinary problems.

- Master thesis: Perturbations propagation in self-organized patterns.

2015-2019 — B.Sc Physics at Universitat de Barcelona

Minor in Fundamental Physics.

- Bachelor thesis: Many-body contributions in water clusters (published in ACS Nano).



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Scientific publications

2026 Computational analysis of Wikipedia editorial activity during the 2024 European Parliament election
D. Abella Bujalance, D. Laniado. Software and data, Zenodo.

2025 Exploring the spatial segmentation of housing markets from online listings
EPJ Data Science 14, 34.

2024 Ordering dynamics and aging in the symmetrical threshold model
New Journal of Physics 26, 013033.

2023 Aging in binary-state models: the Threshold model for complex contagion
Physical Review E 107, 024101.

2023 Many-body contributions in water nanoclusters
ACS Nano 17 (3), 1959-1964.

2023 Unraveling higher-order dynamics in collaboration networks
Preprint, arXiv:2306.17521.

2022 Aging effects in Schelling segregation model
Scientific Reports 12, 19376.

Awards

Nov-2024 Doctoral thesis prize, UIB Chair of Urban Studies (2nd edition)
Prize of 1,500 € for the thesis *Aging and memory effects in social and economic dynamics*.

Collaboration and outreach events

2024 Pint of Science 2024
Talk for a general audience on the physics of social and urban dynamics.

Sep-2023 Associate researcher at Queen Mary University of London (QMUL)
iMOVE research stay with Vincenzo Nicosia to study modeling techniques for spatial segregation.

2022-2023 Poster Party
Member of the Poster Party committee organizing the outreach event at IFISC.

Oct-2022 Conference on Complex Systems 2022
Volunteer and collaborator at the Conference on Complex Systems CCS 2022 in Palma.



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Scientific presentations

Nov-2026	Oral presentation "AI4Bus: Artificial intelligence applied to the metropolitan bus network" at the XII Symposium on Urban Mobility Challenges, CARNET (Barcelona - Spain)
Feb-2026	Presentation of the DEM-Debate results at the project final event, European Parliament (Brussels - Belgium)
Sep-2025	Oral presentation "Exploring Wikipedia community practices during the 2024 European Parliament election" at CCS2025 (Siena - Italy)
May-2025	Presentation "Exploring Wikipedia community practices during the 2024 European Parliament election" at Wiki Workshop 2025 (online)
Feb-2024	Oral presentation "Assessing the real estate spatial segmentation" at CS3-24 (Barcelona - Spain)
July-2023	Oral presentation "Aging in binary state dynamics: Ordering in threshold models" at NetSci23 (Vienna - Austria)
May-2023	Oral presentation "Aging in binary state dynamics: Ordering in threshold models" at CS3-23 (Santander - Spain)
Oct-2022	Oral presentation "Aging in binary-state models: Threshold model of complex contagion" at CCS2022 (Palma - Spain)
Sep-2022	Poster presentation "Aging in binary-state models" at II Summer school on Complex Socio-Technical systems (Palma - Spain)
Aug-2022	Poster presentation "Aging in binary-state models" at X Summer school on Statistical physics of Complex Systems (Palma - Spain)
May-2022	Poster presentation "Aging in Complex contagion" at FISES (Zaragoza - Spain)
Mar-2022	Oral presentation "Aging effects in Schelling segregation model" at COMSOTEC Workshop (Tarragona - Spain)
Oct-2021	Oral presentation "Aging effects in Schelling segregation model" at CCS2021 (Lyon - France)

Supplementary Education

Oct-2026	Agile training by Kairós (at Eurecat)
2023	Participation at the Complexity72h Workshop
2022	Summer school about Artificial Intelligence "Retos de la IA" by AIHUB-CSIC
2022	X Summer school on Statistical physics of Complex Systems by GEFENOL (at IFISC)
2022	II Summer school on Complex Socio-Technical systems by COMSOTEC (at IFISC)
2021	Introduction course to Artificial Intelligence at UBICS